

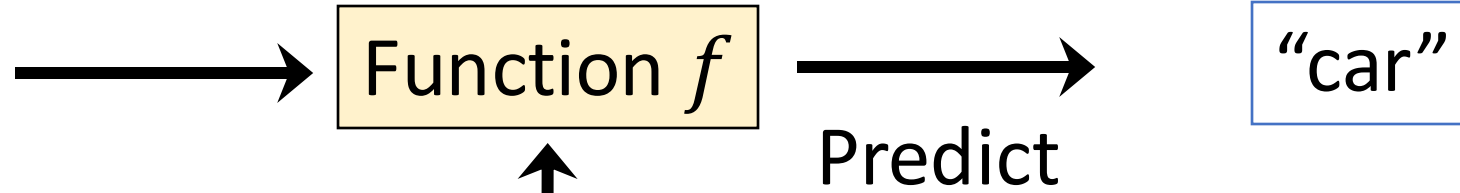
Machine Learning Basics

Erisa Terolli

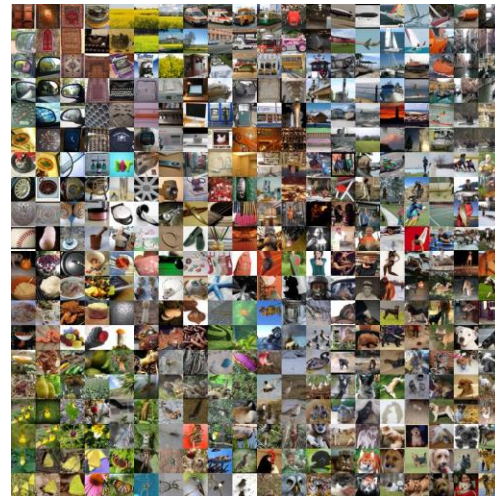
Learning from Data



Future Data



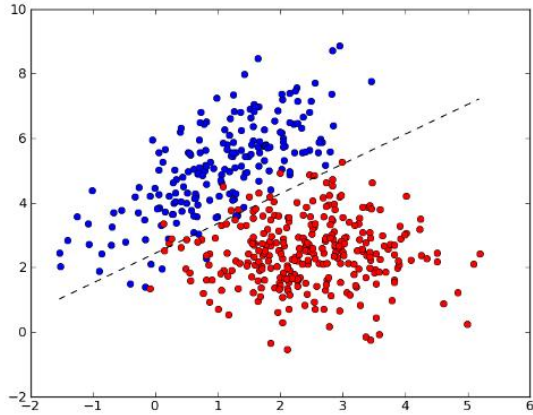
Train



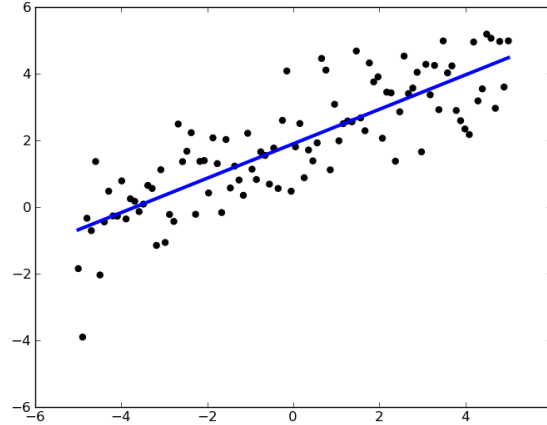
Past Data
(perhaps with **labels**)

Machine Learning Tasks

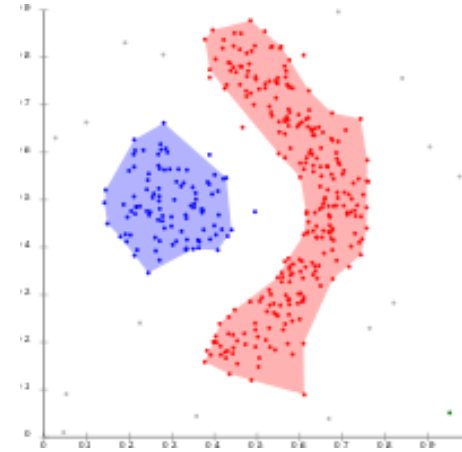
Machine Learning Tasks



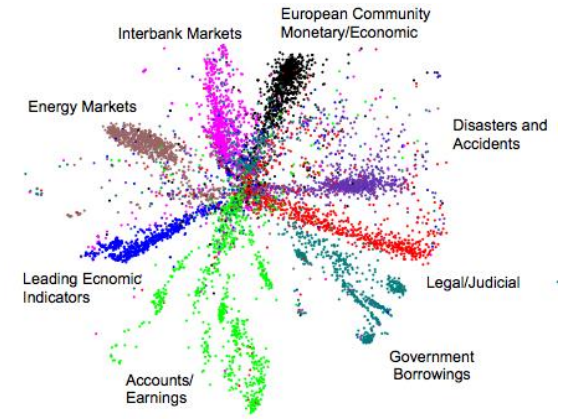
Classification



Regression



Clustering



Dimensionality Reduction

spam
detection

face
recognition

...

Classification: Spam Detection



powered by 



Mail ▾

☐ ▾

COMPOSE

Inbox

Starred

Sent Mail

Drafts

All Mail

Less ▲

Important

Chats

Spam (78)



Shusen ▾

+

☐ ★ 贡蝎

[Delete all spam messages now](#) (messages that have been in Spam more than 30 days will be

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☐ ★ 富珑

业务员为什么有激情，没有业绩？ - shusen 为什么业务员经常申请政策，但就是没有业绩？ 为

☐ ★ Western Dental & Orthodo.

Happy Halloween, We Are Open October 31 🧡 - Logo 866-469-5955 Come celebrate with u

☐ ★ kqaqy2

Conference to-shusen - P80806PN840XJRF20V8HLFFP0V8V6206088J8P8BX0JX4NFDB4I

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Why work stress isn't all bad, and other career trends this week - New: Career tips, news,

Classification: Spam Recognition

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We invite you to submit your manuscript(s) for publication. The journals include research papers, review articles, technical projects and short communications containing new insight into any aspect of the covered scope of the journal. Our objective is to inform authors of the decision on their manuscript(s) within weeks of submission. After acceptance, the paper will be published in the current issue immediately.

Keywords: English, Literature, Science, Economics, Engineering, Management, Agriculture, Horticulture, Environment

[International Journal of Advanced Engineering Research and Science \(IJAERS\)](#) ISSN: 2456-1908(O) | 2349-6495 (P)

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Thomson Reuters ResearcherID: [P-3738-2015](#)

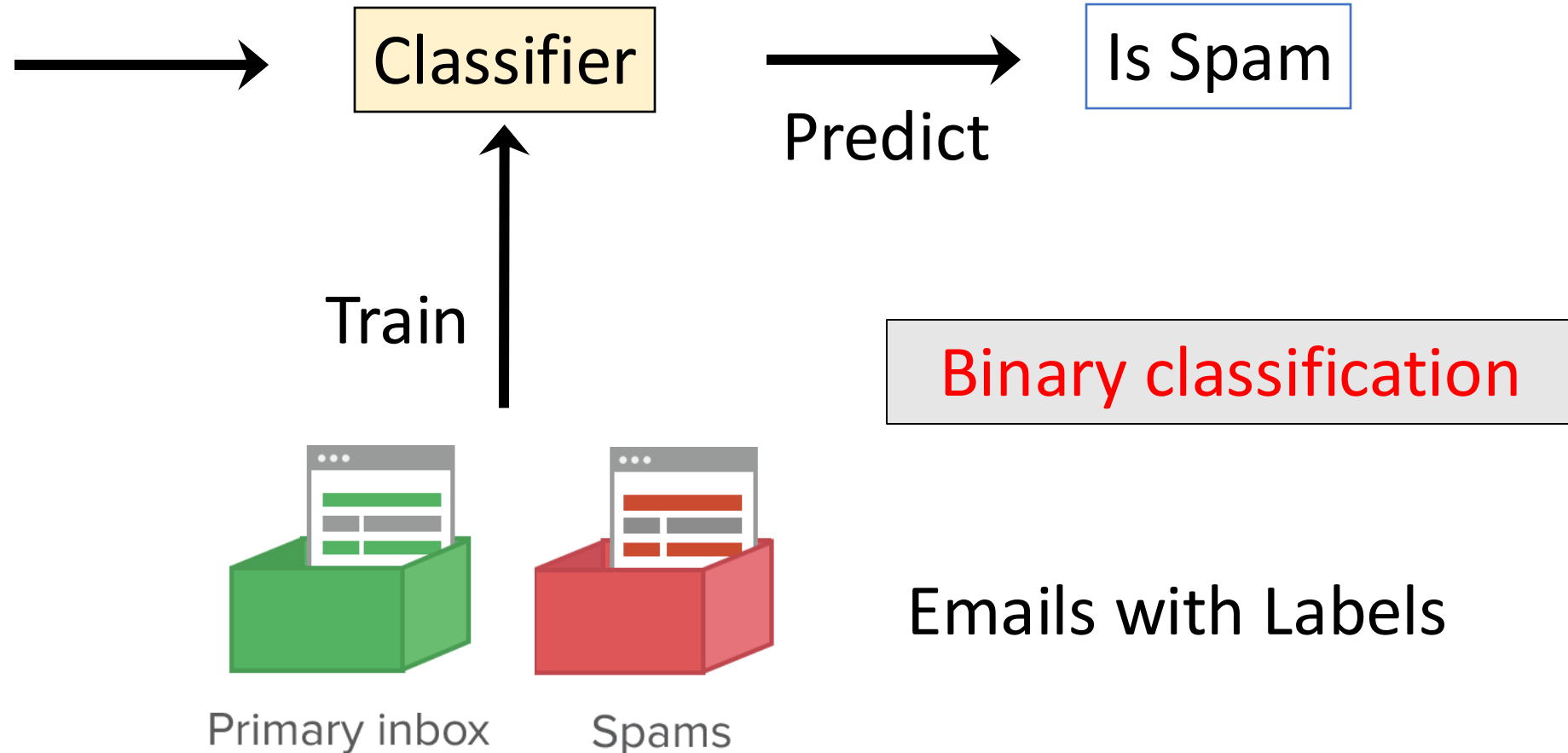
Impact Factor: 4.192, SJIF: 4.072, IBI: 3.2, PIF: 2.465, ISRA-JIF: 1.317,

Website: <http://www.ijaers.com>

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[International Journal of English, Literature and Science \(IJELS\)](#)
ISSN: 2456-7620

New Email



Classification: Spam Recognition

Inbox

1,195

★

Starred

🕒

Snoozed

➤

Sent

📄

Drafts

✉

All Mail

^

Less

➤

Important

💬

Chats

⚠

Spam

141

🗑

Trash

📁

Primary

👤

Social

🏷

Promotions

★

Shutterfly

Ad

📄

ENJOY UP TO 50% OFF - Woven Photo Blankets: Upload Your Own Design W

★

Canvas Champ

Ad

📄

Best Online Photo Printing Service - Best Online Photo Printing Service Trus

☐

★

»»

Walmart Fashion

Fill their holidays with festive kids' fashion 🧸 - Find last-minute gifts, party

☐

★

»»

eBay

Spotted: Falling prices that have got us excited! - Wow, so much newness! \$

☐

★

»»

Costco Wholesale

Dinner is Served! Same-Day Delivery on all Your Holiday Dinner Essentials.

☐

★

»»

IKEA USA

Friend, in need of a last minute gift? - Give an IKEA Gift Card View in Browse

☐

★

»»

Walmart

LAST CHANCE to save up to 56% 🙌🙌🙌 - Here's your final chance to get 1

☐

★

»»

Costco Wholesale

LAST HOURS to order in time for Christmas delivery. - Gift ideas - Mac, PCs

☐

★

»»

Walmart

HURRY! Last call for free 2-day shipping - Order by Dec. 20 at 2pm for delive

☐

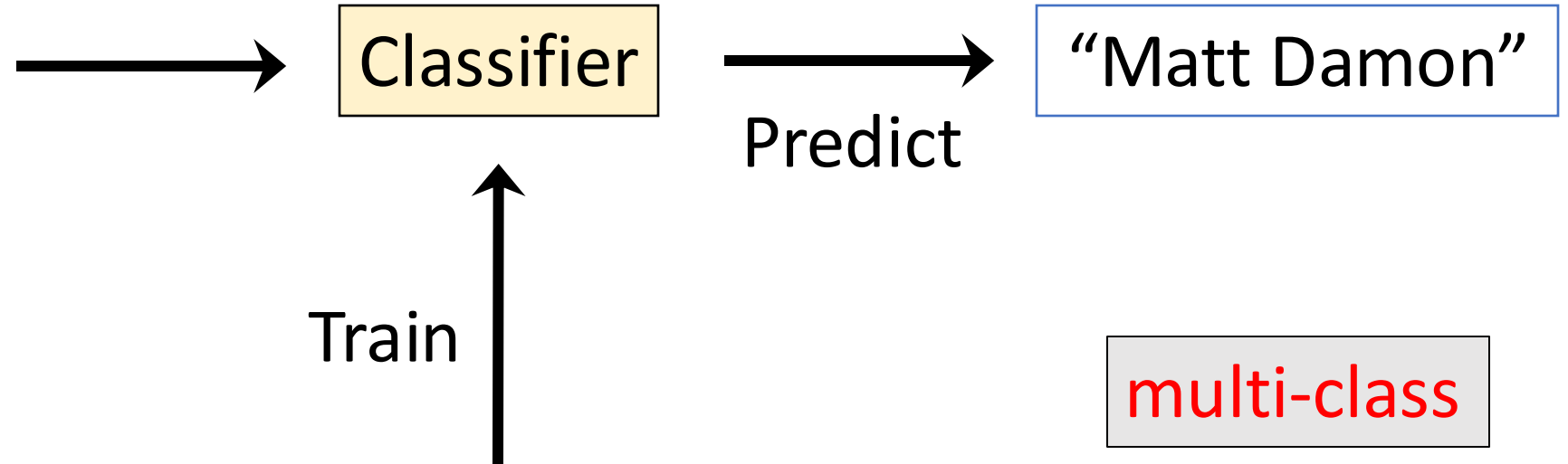
★

»»

Hilton Honors

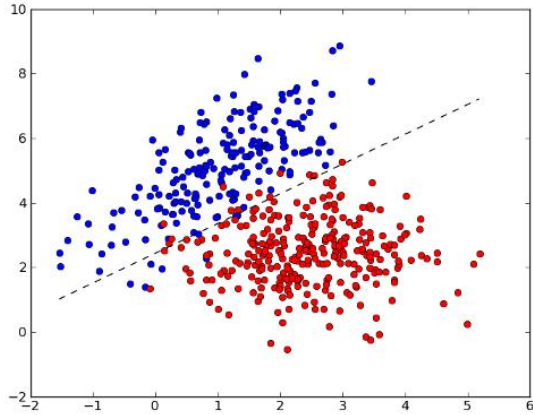
'Tis the season for unlimited Points - register now - Start planning your 201

Classification: Face Recognition

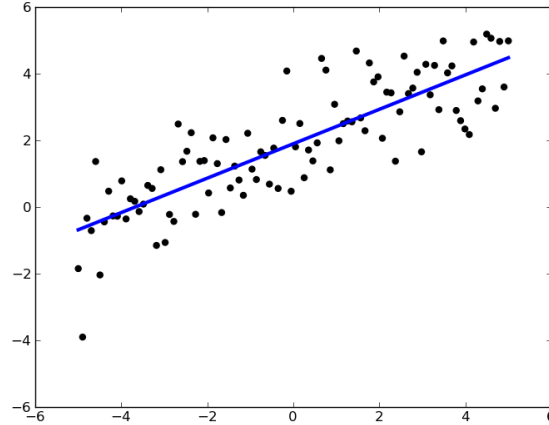


Faces & Names

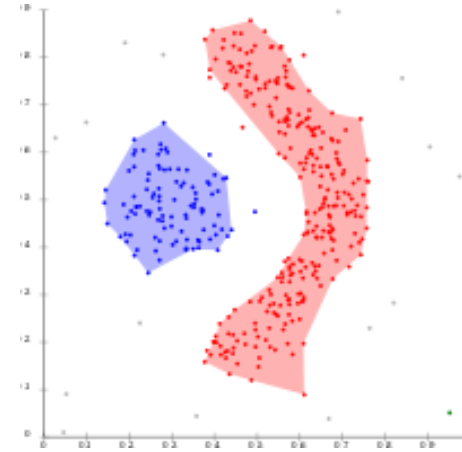
Machine Learning Tasks



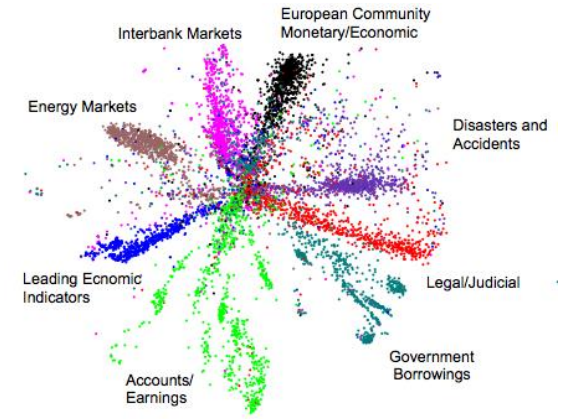
Classification



Regression



Clustering



Dimensionality
Reduction

Regression: Housing Price



Features of a House



Regressor



Predict

Price = \$0.5M

Train



label (target)

features

Price

Beds

Baths

Square Feet

Miles to Resort

Miles to Base

Acres

Cars

Years Old

DoM

Selling price of the property (\$1,000)

Number of bedrooms in the house

Number of bathrooms in the house

Size of the house in square feet

Miles from the property to the downtown resort area

Miles from the property to the base of the ski resort's mountain

Lot size in number of acres

Number of cars that will fit into the garage

Age of the house, in years, at the time it was listed

Number of days the house was on the market before it sold

Regression VS Classification

- **Regression: labels are ordered**
- Example:

House Prices:

\$324K

<

\$521K

<

\$1.2M



Regression VS Classification

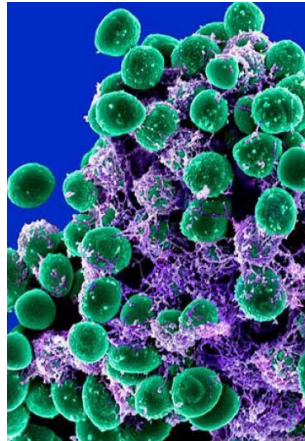
- Regression: labels are ordered
- **Classification: labels are categorical**
- Example:

Categories:

Class 1



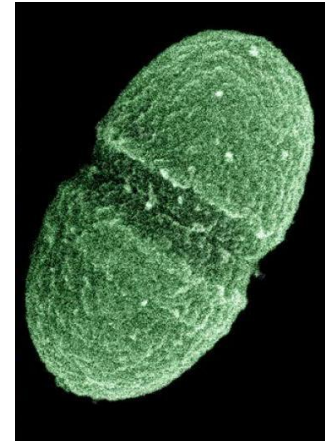
Class 2



Class 3



Class 4



Regression VS Classification

- Regression: labels are ordered
- **Classification: labels are categorical**
- Example:

Categories:

Class 1

Class 2

Class 3

Class 4

“Class 1” and “Class 4” are categories. They are not ordered!

Class 1



Class 4

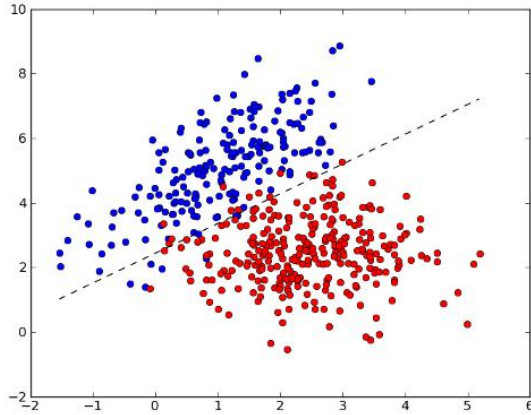
Regression VS Classification

Problem: Wine Quality Prediction.

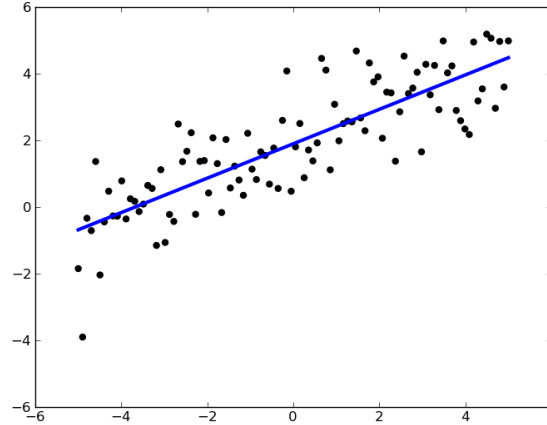
- **Features:** Measurements of chemicals in the wine.
- **Targets:** Outstanding, very good, good, fair, not recommended.
- Is wine quality prediction a **regression** or **classification** problem?



Machine Learning Tasks

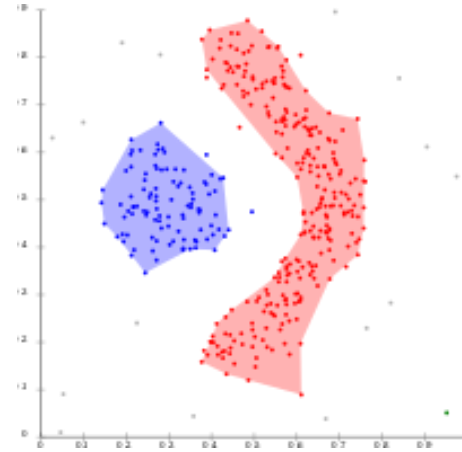


Classification



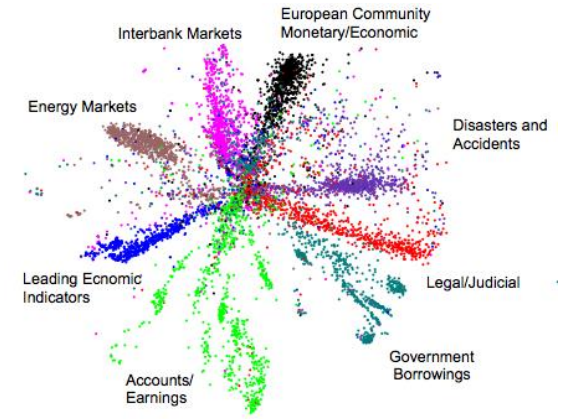
Regression

Supervised Learning



Clustering

Unsupervised Learning



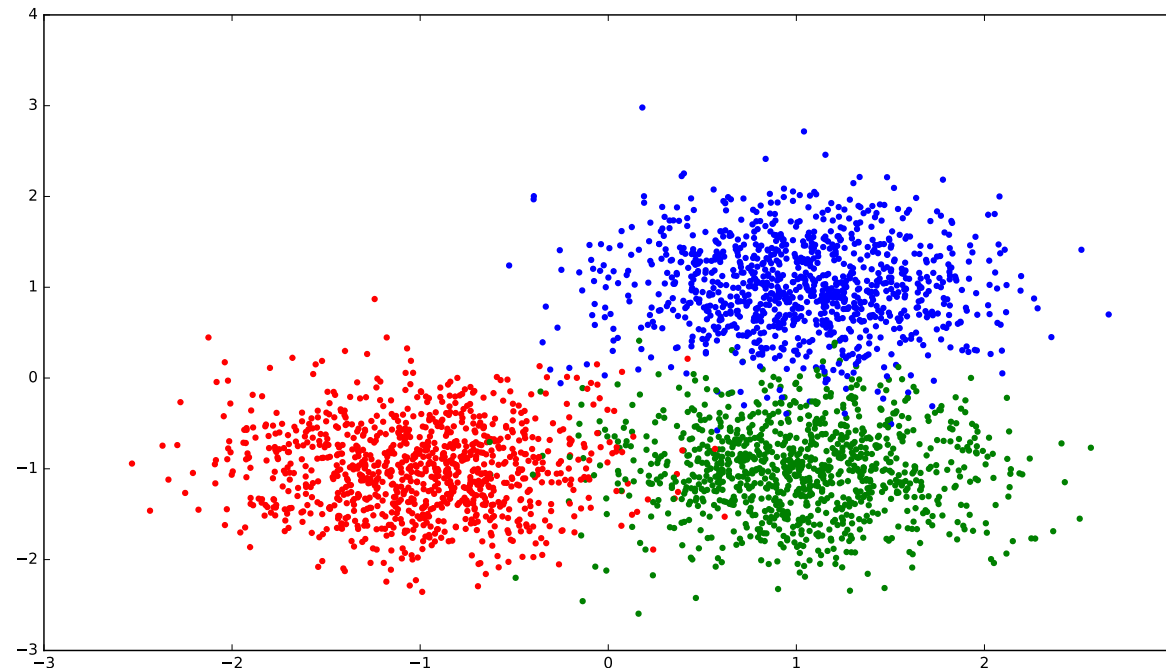
Dimensionality Reduction

Unsupervised Learning
Supervised Learning

Clustering: Unsupervised Learning

Input: vectors $\mathbf{x}_1, \dots, \mathbf{x}_n$ and number of clusters k ($\ll n$).

Output: predicted classes $y_1, \dots, y_n \in \{1, 2, \dots, k\}$.



Clustering: Unsupervised Learning

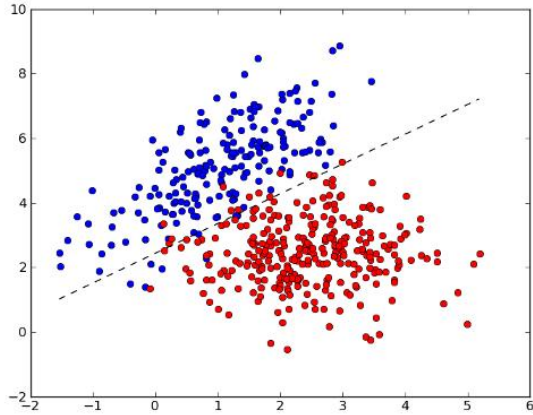
Supervised learning : learn function from **labeled** training data.

- Classification and regression are supervised learning tasks.

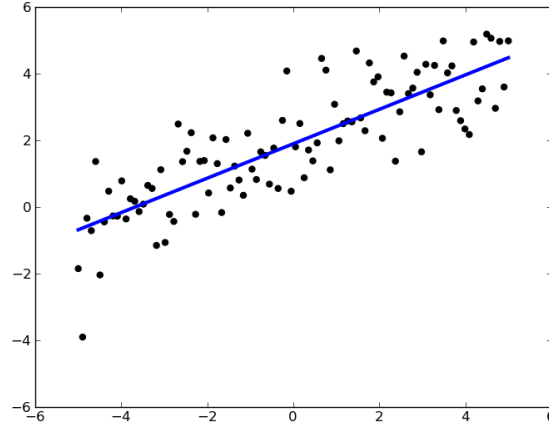
Unsupervised learning : draw inferences from datasets consisting of data **without label**.

- Clustering is an unsupervised learning task.

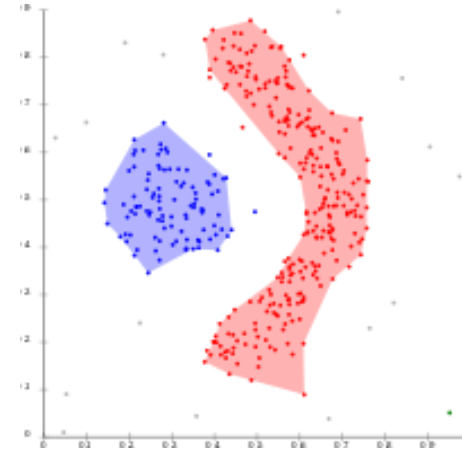
Machine Learning Tasks



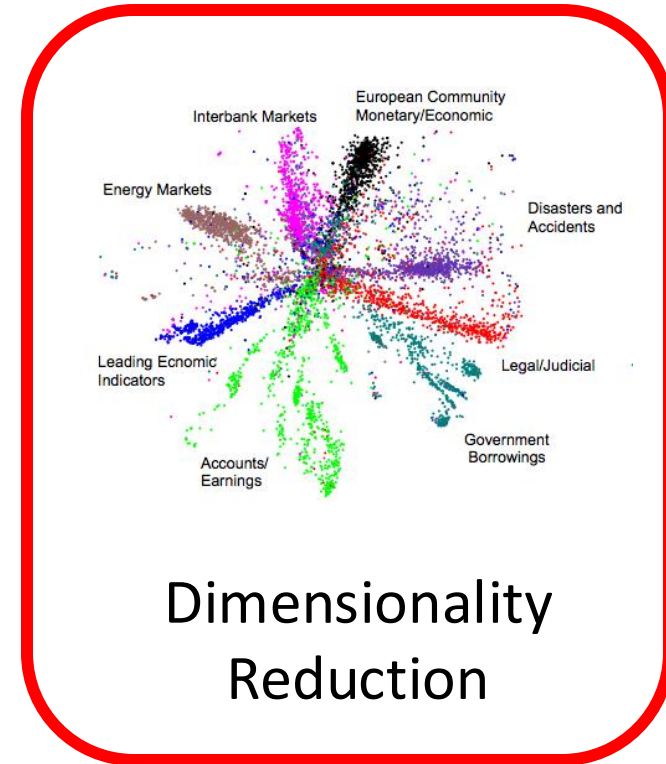
Classification



Regression

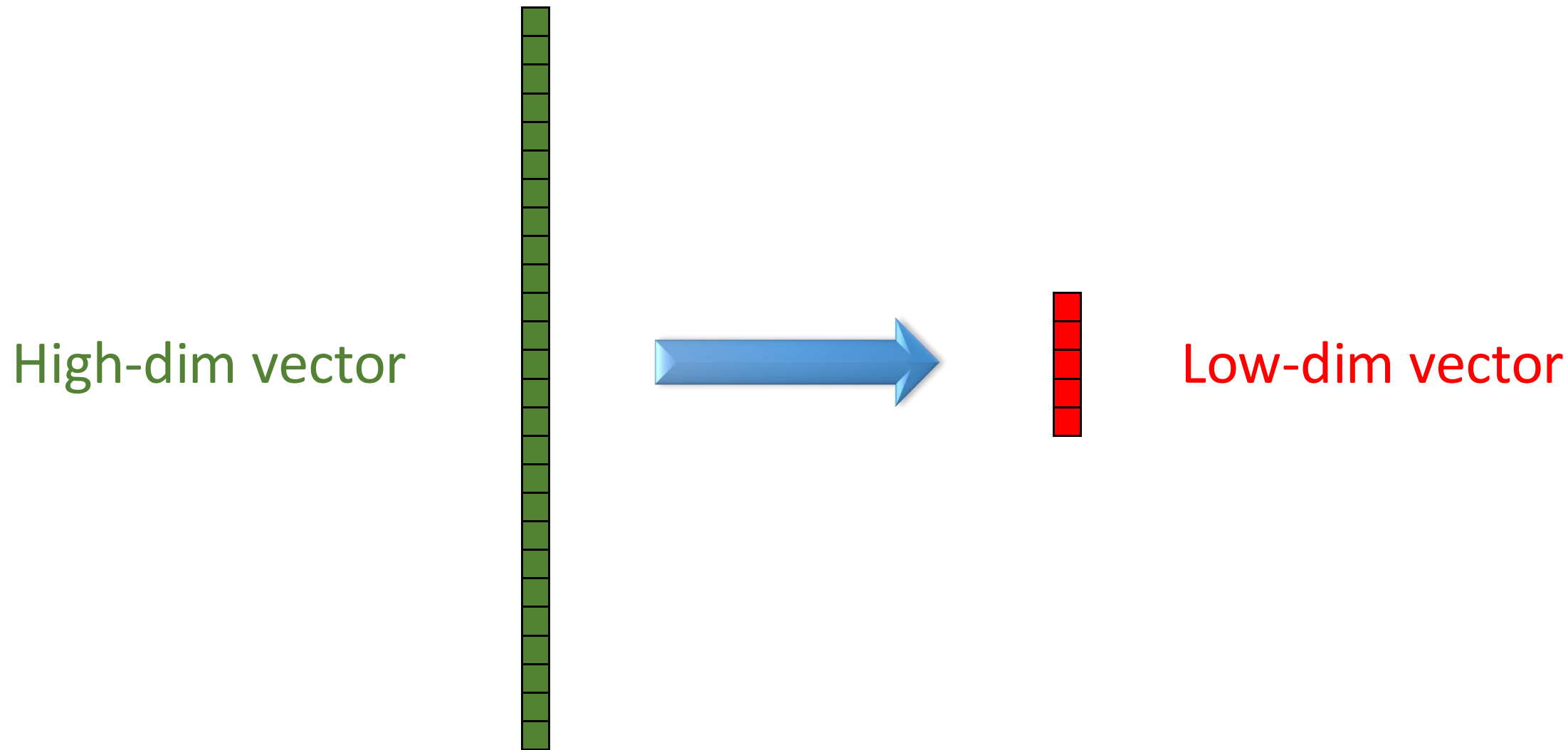


Clustering



Dimensionality
Reduction

Dimensionality Reduction



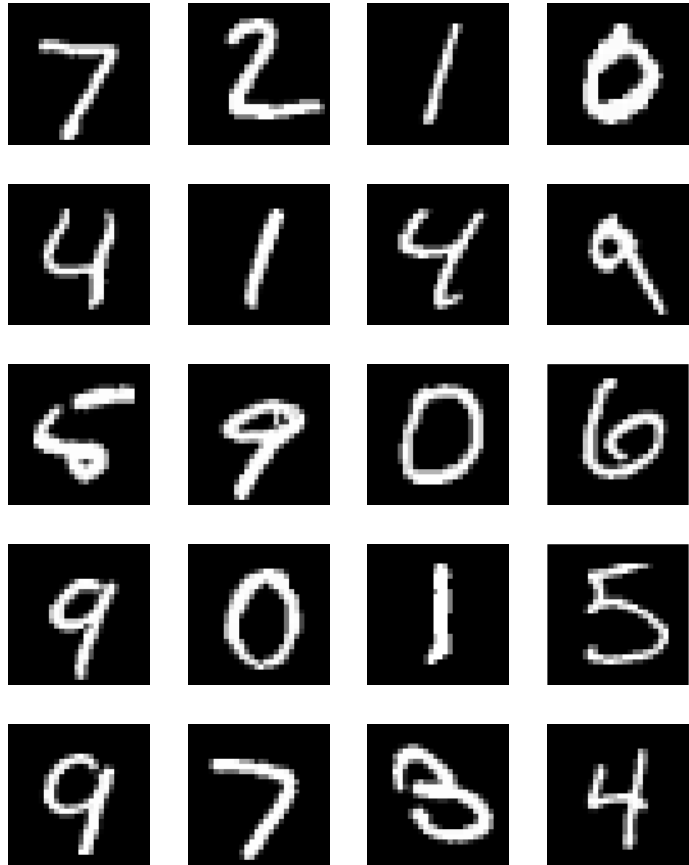
Dimensionality Reduction

- Applications
 - Visualization and analysis
 - Data processing (to make downstream ML more efficient or accurate)

Dimensionality Reduction

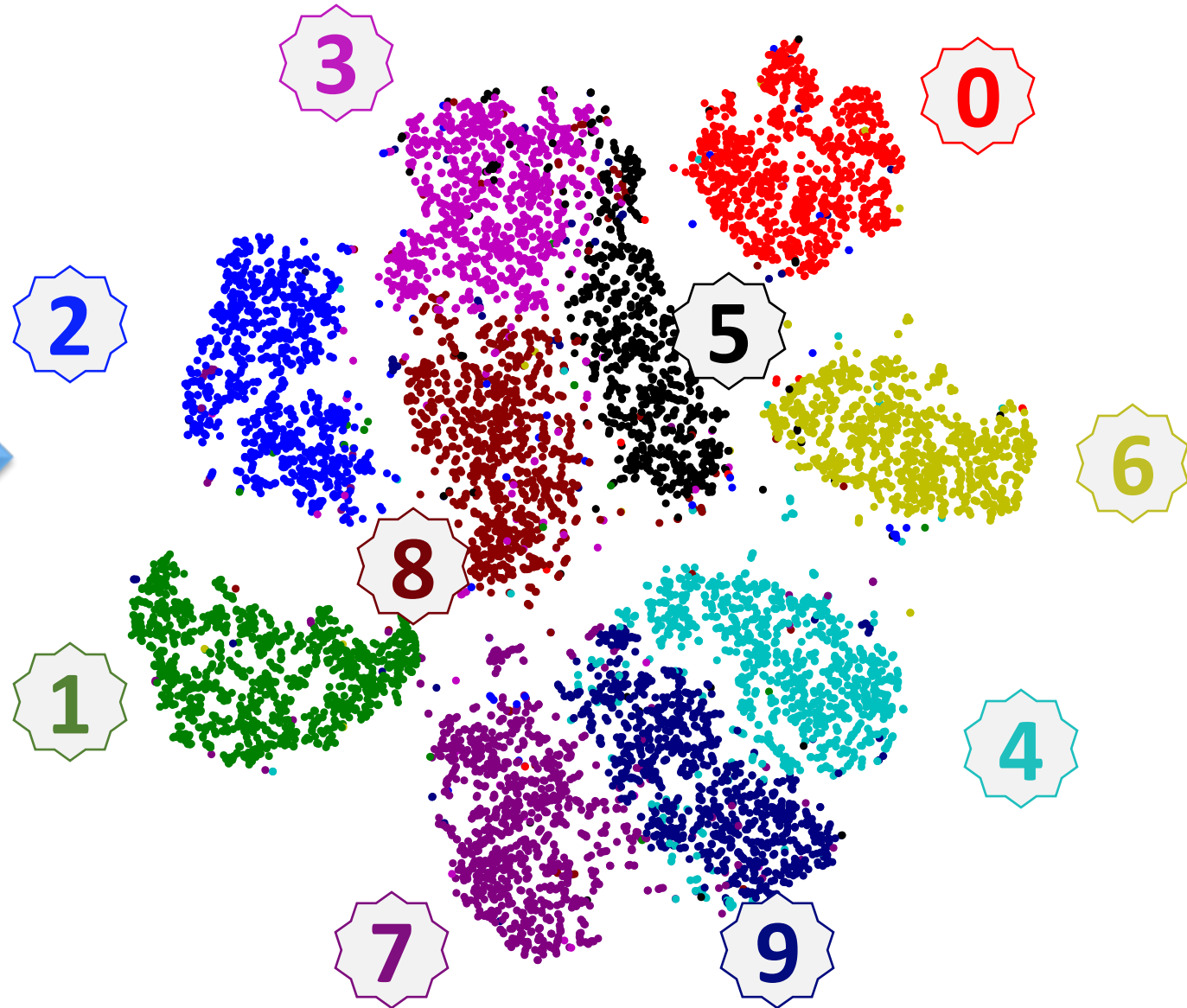
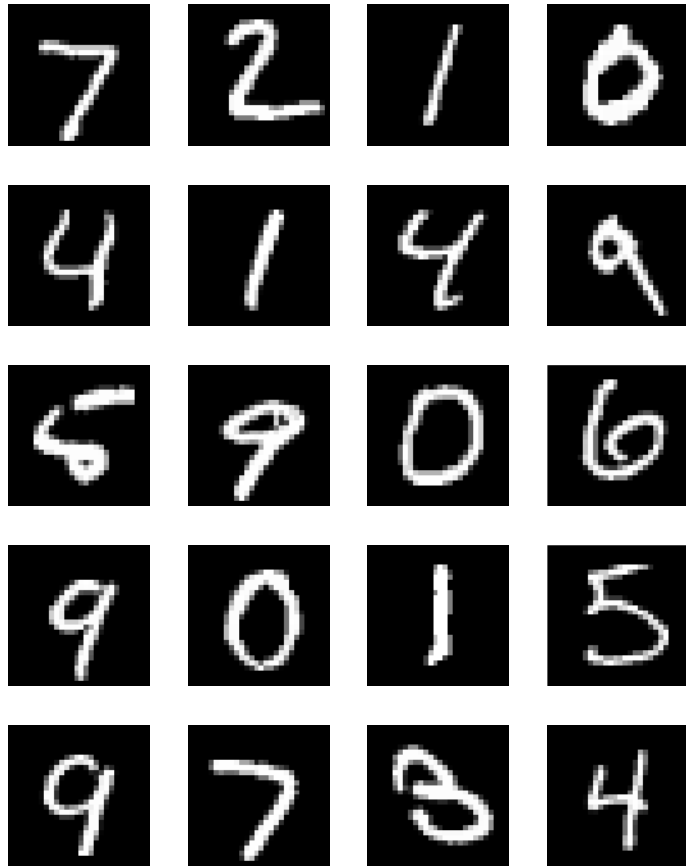
- Applications
 - Visualization and analysis
 - Data processing (to make downstream ML more efficient or accurate)
- Methods
 - Principal component analysis (PCA)
 - Kernel PCA
 - Manifold learning
 - Autoencoder
 - Linear discriminant analysis (LDA)

Autoencoder for Hand-Written Digit Data



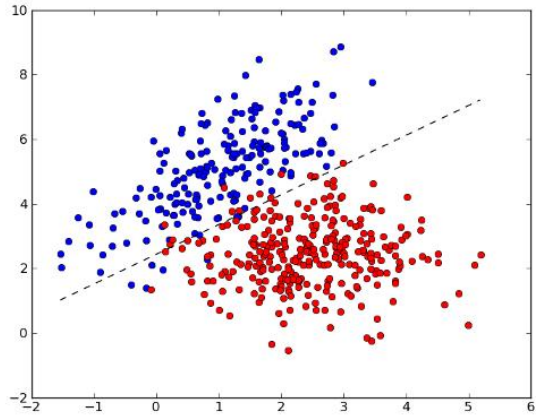
60K images (size 28×28)

Autoencoder for Hand-Written Digit Data

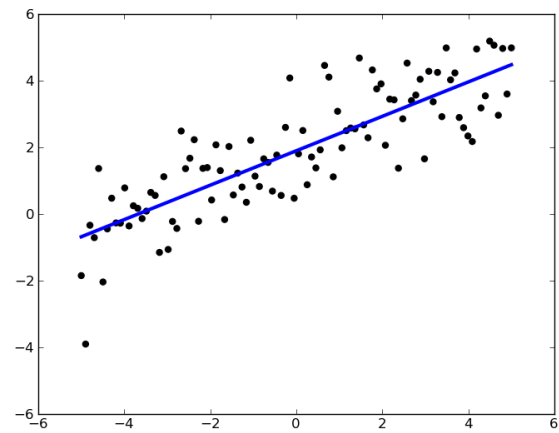


60K images (size 28×28)

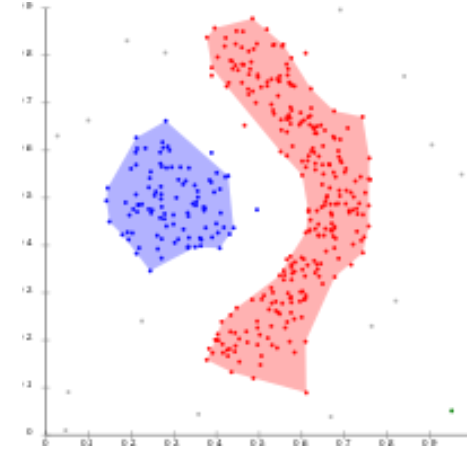
Machine Learning Tasks



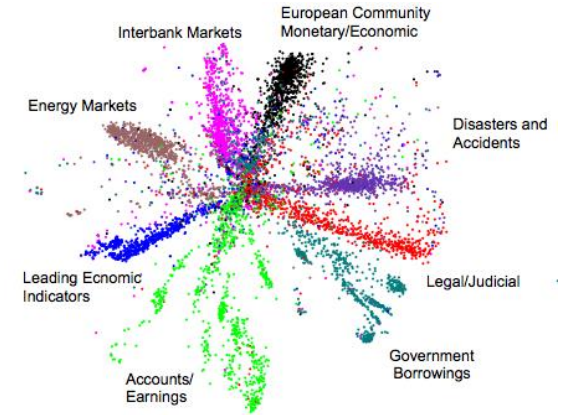
Classification



Regression

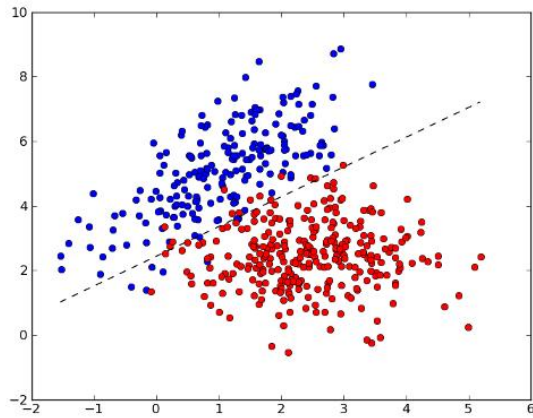


Clustering

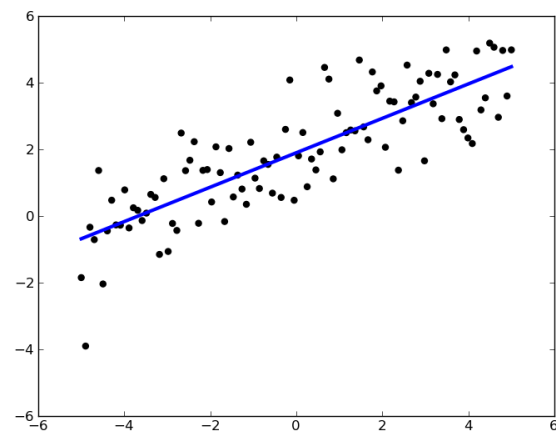


Dimensionality
Reduction

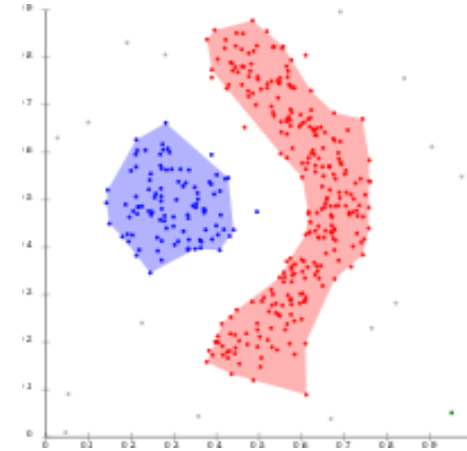
Machine Learning Tasks



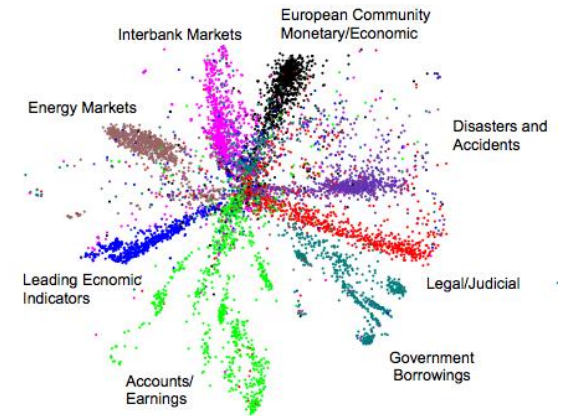
Classification



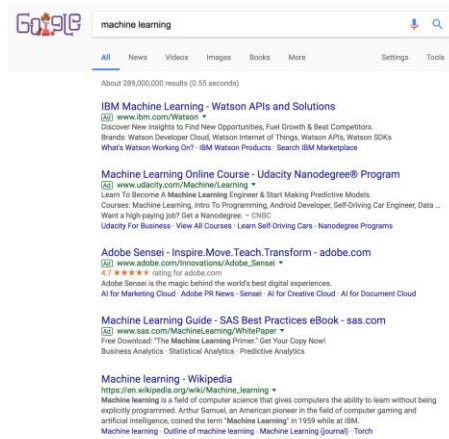
Regression



Clustering



Dimensionality Reduction



Ranking



Recommendation



Reinforcement Learning

Tasks, Methods, & Algorithms

Tasks

(or problems)

Regression

Classification

Clustering

Dim Reduction

Tasks (or problems)

Regression

Classification

Clustering

Dim Reduction

Methods (or models)

Neural Networks

SVM

Logistic Regression

Decision Tree

Nearest Neighbor

Tasks (or problems)

Regression

Classification

Clustering

Dim Reduction

Methods (or models)

Neural Networks

SVM

Logistic Regression

Decision Tree

Nearest Neighbor

Algorithms

Gradient Descent (GD)

Stochastic GD

Coordinate Descent (CD)

Dual CD